The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 August 2008.

INCH-POUND

MIL-PRF-19500/115M
14 May 2008
SUPERSEDING
MIL-PRF-19500/115L
5 July 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE REGULATOR, TYPES
1N3821A THROUGH 1N3828A, 1N3016B THROUGH 1N3051B,
1N3821A-1 THROUGH 1N3828A-1, 1N3016B-1 THROUGH 1N3051B-1,
1N3821AUR-1 THROUGH 1N3828AUR-1, 1N3016BUR-1 THROUGH 1N3051BUR-1,
PLUS C- AND D- TOLERANCE SUFFIX,
JAN, JANTX, JANTXV, AND JANHC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 1 W, silicon, voltage regulator diodes with voltage tolerances of 5 percent, 2 percent, and 1 percent. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500. One level of product assurance is provided for die.

1.2 Physical dimensions. See figures 1 (DO-13), 2 (DO-41), 3 (DO-213AB), 4, and 5 (for JANHC).

1.3 Maximum ratings. Maximum ratings are as shown in maximum test ratings (see 3.8) herein and as follows: $-55°C \leq T_{op} \leq +175°C$; $-55°C \leq T_{STG} \leq +175°C$.

| Type | $P_{TL}$ | $T_L$ | $T_{EC}$ | $P_{TPCB}$ |
|---|---|---|---|---|
| | W | °C | °C | W |
| DO-13, DO-41 | 1.0 (1) | +95 | | 1 |
| DO-213AB | 1.0 (2) | | +125 | |

(1) L = .375 inch (9.53 mm). Both ends of case or diode body to heat sink at L = .375 (9.53 mm). (Derate $I_Z$ to 0.0 mA dc at $P_{TL}$ = +175°C).

(2) Derate to 0 at $P_{TEC}$ = +175°C.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in primary test ratings (see 3.9) herein and as follows: 3.3 V dc ≤$V_Z$ ≤200 V dc. A and B suffix devices are 5 percent voltage tolerance. C suffix devices are 2 percent voltage tolerance. D suffix devices are 1 percent voltage tolerance.

| Type | $R_{\theta JL}$ (1) | $R_{\theta JEC}$ (2) | $R_{\theta JA}$ (3) |
|---|---|---|---|
| | °C/W | °C/W | °C/W |
| DO-13 | 80 | | |
| DO-41 | 80 | | |
| DO-213AB | | 50 | |

(1) L = .375 inch (9.53 mm).
(2) Junction to end-caps.
(3) See figures 6, 7, and 8 for derating curves. $T_A$ = +75°C for both axial and MELF (US) on printed circuit board (PCB), PCB = FR4 .0625 inch (1.59 mm) 1-layer 1-Oz Cu, horizontal, still air, pads (US) = .067 inch (1.70 mm) x .105 inch (2.67 mm); pads (axial) = .092 inch (2.34 mm) diameter, strip = .030 inch (0.762 mm) x 1 inch (25.4 mm) long, axial lead length L ≤.187 inch (≤4.76 mm); $R_{\theta JA}$ with a defined thermal resistance condition included is measured at $I_Z$ = as defined in test ratings herein.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil.or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

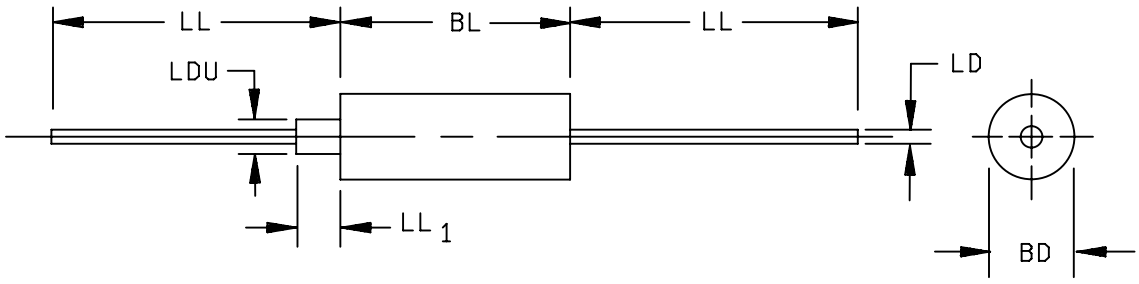
3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).



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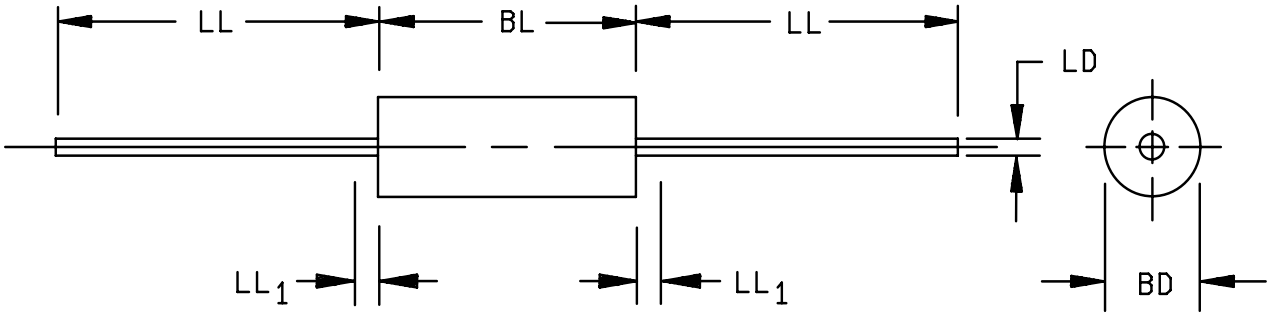

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BD | .215 | .265 | 5.46 | 6.73 | 3 |
| BL | .195 | .350 | 4.96 | 8.89 | |
| LD | .026 | .035 | 0.66 | 0.89 | 4 |
| LDU | | .110 | | 2.79 | |
| LL | 1.000 | | 25.40 | | |
| $LL_1$ | | .21 | | 5.33 | |

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. Cathode lead shall be electrically connected to the case. If tubulation is used, it shall be on the anode end.
5. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 1. Physical dimensions types 1N3821A, C, D through 1N3828A, C, D and 1N3016B, C, D through 1N3051B, C, D (DO-13).



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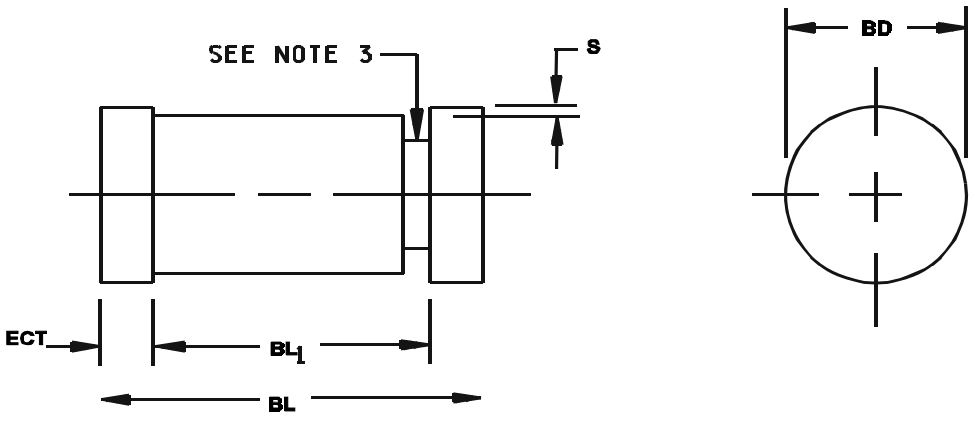

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BD | .080 | .107 | 2.03 | 2.72 | 3 |
| BL | .160 | .205 | 4.06 | 5.21 | 3 |
| LD | .028 | .034 | 0.71 | 0.86 | |
| LL | 1.000 | | 25.40 | | |
| $LL_1$ | | .50 | | 12.7 | 4 |

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone lead, diameter may vary to allow for lead finishes and irregularities other than heat slugs.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions types 1N3821A-1, C-1, D-1 through 1N3828A-1, C-1, D-1 and 1N3016B-1, C-1, D-1 through 1N3051B-1, C-1, D-1 (DO-41).



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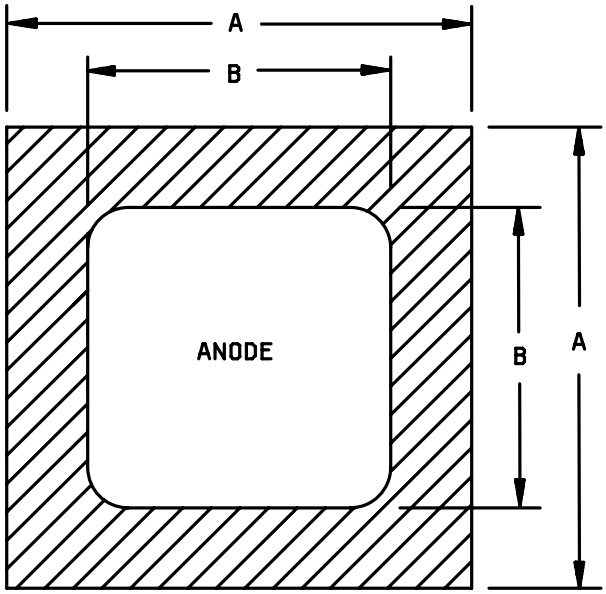

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BD | .094 | .105 | 2.39 | 2.67 |
| $BL_1$ | .159 (Ref.) | | 4.04 (Ref.) | |
| BL | .189 | .205 | 4.80 | 5.21 |
| ECT | .014 | .022 | 0.360 | 0.560 |
| S | .001 | | 0.030 | |

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. Gap not controlled, shape of body and gap not controlled.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions of surface mount family types 1N3821AUR-1, CUR-1, and DUR-1 through 1N3828AUR-1, CUR-1, DUR-1 and 1N3016BUR-1, CUR-1 and DUR-1 through 1N3051BUR-1, CUR-1 and DUR-1 (DO-213AB).



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A Version

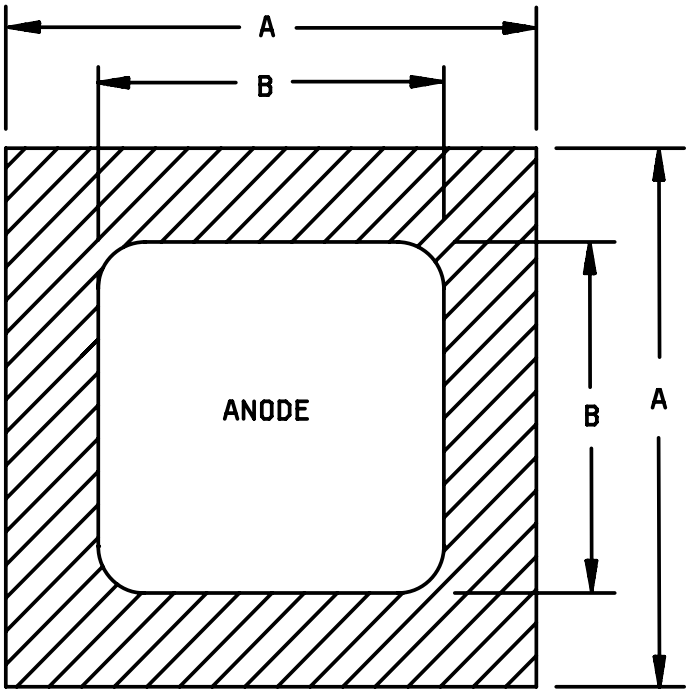
| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .035 | .039 | 0.89 | 0.99 |
| B | .031 | .033 | 0.79 | 0.84 |

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. The physical characteristics of the die thickness are .010 ±002 (0.25 mm). Metallization is: Top (anode) - Al, back (cathode) - Au. Al thickness = 25,000Å minimum, Au thickness = 4,000Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions JANHCA die.



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BACKSIDE IS CATHODE

B Version

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .035 | .039 | 0.89 | 0.99 |
| B | .027 | .031 | 0.68 | 0.79 |

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. The physical characteristics of the die thickness are .012 ±002 (0.30 mm). Metallization is:
   Top (anode) - Al, back (cathode) - Au. Al thickness = 40,000Å minimum, Au thickness = 5,000Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 5. Physical dimensions JANHCB die.



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3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 (DO-13), 2 (DO-41), 3 (DO-213AB), and figures 4 and 5 for (JANHC).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-PRF-19500 where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. All devices shall be in accordance with the requirements of MIL-PRF-19500 and as follows.

3.4.2.1 Dash one construction. Dash one (-1) diodes shall be of metallurgically bonded double plug construction, (category I, II, and III in accordance with MIL-PRF-19500).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking of UR version devices. For UR version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.5.2 Polarity. For dash one or UR dash one, the polarity shall be indicated with a contrasting color band to denote the cathode end or alternately with a minimum of three contrasting color dots spaced evenly around the periphery at the cathode end.

3.6 Selection of tight tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, or JANTXV devices, which have successfully completed all applicable screening, and groups A, B, and C testing as five (5) percent tolerance devices. All sublots of C and D suffix devices shall pass table I, subgroup 2, at tighter tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purpose to establish correlation. For C and D tolerance levels, $T_L$ = 30 $\pm$2°C at .375 inches (9.53 mm) from body or equivalent.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 (see table I, II, and III herein).

3.9 Maximum and primary test ratings. Maximum and primary test ratings for voltage regulator diodes are specified in table III herein.

3.10 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.



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4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC devices. JANHC devices shall be qualified in accordance with MIL-PRF-19500.

4.2.3 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.3 Screening (JAN, JANTXV, and JANTX levels only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see appendix E, table E-IV of MIL-PRF-19500) | Measurement |
|---|---|
| | JANTX and JANTXV levels |
| 3a | Temperature cycling |
| (1) 3c | Thermal impedance (see 4.3.2) |
| 7a<br>7b | Not applicable<br>Optional |
| 9 | Not applicable |
| 11 | $I_{R1}$ and $V_Z$ |
| 12 | See 4.3.3 |
| (2) 13 | Subgroup 2 of table I herein.<br>$\Delta I_{R1} \leq$ 100 percent of initial reading or 50nA dc, whichever is greater<br>$\Delta V_Z \leq \pm$ 2 percent of initial reading. |
| 14a<br>(3) 14b | Applies to DO-13 devices only<br>Required |

(1) Thermal impedance may be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test. (Applicable to –1 and UR-1 devices only).
(2) PDA = 5 percent for screen 13, applies to $\Delta I_{R1}$, $\Delta V_Z$. Thermal impedance ($Z_{\theta JX}$) is not required in screen 13.
(3) For clear glass diodes, the hermetic seal (gross leak) may be performed at anytime after temperature cycling.

4.3.1 Screening (JANHC). Screening of JANHC die shall be in accordance with MIL-PRF-19500.

4.3.1.1 JAN testing. Temperature cycling and thermal impedance testing shall be performed in accordance with JANTX requirements.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 as applicable of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table II, group E, subgroup 4 herein.

4.3.3 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.5): $I_{Z(min)}$ = column 8 of table III; $T_A$ = 75°C maximum. Test conditions in accordance with method 1038 of MIL-STD-750, condition B. Adjust $I_Z$ or $T_A$ to achieve the required $T_J$. $T_J$ = 125°C minimum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroups 2 and 4 herein.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | -55°C to +175°C, 25 cycles. |
| B2 | 4066 | See 4.5.1. |
| B3 | 1027 | $I_Z = I_{ZM}$ column 8 of table III (min); Adjust Iz or $T_A$ to achieve $T_J$ = 150°C (min). |
| B4 | 2101 | Decap analysis scribe and break only. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroups 2 and 4 herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Terminal strength: Test condition A; weight = 4 lbs; t = 15 seconds.<br>Terminal strength: Test condition E. (Terminal strength not required for UR-1 devices.) |
| C3 | | Applies to DO-13 devices only. |
| C5 | 4081 | See 4.5.7. |
| C6 | 1026 | $I_Z = I_{ZM}$ column 8 of table III (min); Adjust Iz or $T_A$ to achieve $T_J$ = 150°C (min). |
| C8 | 4071 | $I_Z = I_Z$ column 5 of table III; $T_A$ = +25°C ±5°C; $T_2$ = +125°C ± 5°C; $\alpha V_Z$ = column 15 of table III; 22 devices, c = 0. |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Surge current ($I_{ZSM}$). The peak currents shown in column 10 of table III shall be applied in the reverse direction and these shall be superimposed on the current ($I_Z = I_{Z1}$) (column 5 of table III) a total of five (5) surges at 1 minute intervals. Each individual surge shall be one-half square-wave-pulse of 1/120 second duration or an equivalent one-half sine wave with the same effective rms current.

4.5.2 Regulator voltage measurements. The test current shall be applied until thermal equilibrium is attained (90 seconds maximum) prior to reading the breakdown voltage. For this test, the surface mount device shall be mounted at the end-caps and the axial leaded device shall be suspended by its leads with mounting clips whose inside edge is located at .375 inch (9.53 mm) from the body and the mounting clips shall be maintained at a temperature of +25°C +8°C,-2°C . This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.



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4.5.3 Temperature coefficient of regulator voltage ($\alpha V_Z$). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature as specified in 4.4.3, subgroup C8.

4.5.4 Voltage regulation $V_Z$ (reg). Voltage regulation shall be determined by the difference of the regulator voltage measured at different currents as specified in table I, subgroup 7. Both tests shall be performed at thermal equilibrium. This $\Delta V_Z$ shall not exceed column 9 of table III.

4.5.5 Free air burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the $I_Z$(min) described in 4.3.3 and that the minimum applied voltage, where applicable, is maintained through-out the burn-in period. Use method 3100 of MIL-STD-750 to measure $T_J$.

4.5.6 For initial qualifications and requalifications. Read and record data in accordance with table II herein and shall be included in the qualification report.

* 4.5.7 Thermal resistance. Thermal resistance measurement shall be in accordance with method 4081 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, and $t_H$. Measurement delay time $t_{MD}$ = 70 μs maximum. See MIL-PRF-19500, table E-IX, subgroup 4. Forced moving air or draft shall not be permitted across the device during test.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits 2/ | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Forward voltage | 4011 | $I_F$ =200 mA dc | $V_F$ | | 1.2 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = column 11 of table III. | $I_{R1}$ | | Col. 12 | µA dc |
| Regulator voltage (see 4.5.2) | 4022 | $I_{Z1} = I_Z$ = (column 5 of table III). | $V_Z$ | Col. 3 | Col. 4 | V dc |
| Thermal impedance | 3101 | See 4.3.2 (-1 device only). | $Z_{\theta JX}$ | | | °C/W |
| Subgroup 3 | | | | | | |
| High-temperature operation | | $T_A$ = +150°C | | | | |
| Reverse current (-1 device only) | 4016 | DC method: $V_R$ = column 11 of table III. | $I_{R2}$ | | Col. 14 | µA dc |
| Subgroup 4 | | | | | | |
| Small-signal reverse breakdown impedance | 4051 | $I_Z$ = (column 5 of table III). $I_{sig}$ =10 percent of $I_Z$. | $Z_{ZT}$ | | Col. 6 | ohms |
| Small-signal knee impedance | 4051 | $I_{ZK}$ = (column 16 of table III). $I_{sig}$ =10 percent of $I_{ZK}$. | $Z_{ZK}$ | | Col. 7 | ohms |
| Subgroups 5 and 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Voltage regulation (see 4.5.4) | | $I_Z$ = 10 percent of column 8 of table III (current 1). $I_Z$ = 50 percent of column 8 of table III (current 2). | $V_Z$(reg) | | Col. 9 | V dc |

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table IV herein.



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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection 1/ | MIL-STD-750 | | Qualification conformance inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices, c = 0 |
| Temperature cycling | 1051 | 500 cycles. | |
| Thermal shock | 1056 | 500 cycles. | |
| Hermetic seal | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 and 4. | |
| Subgroup 2 | | | 45 devices, c = 0 |
| Intermittent life | 1037 | 6,000 cycles. | |
| Electrical measurements | | See table I, subgroup 2 and 4. | |
| Subgroup 3 | | | 3 devices, c = 0 |
| Decap analysis | 2101 | Cross section or scribe and break. Separate samples shall be used for each test. | |
| Subgroup 4 | | | |
| Thermal impedance curves (-1 devices only) | | See MIL-PRF-19500. | |
| Subgroup 5 and 6 | | | |
| Not applicable | | | |
| Subgroup 8 | | | |
| Resistance to glass cracking (-1 devices only) | 1057 | Condition B. Cool down after solder immersion is permitted. Test until failure occurs on all devices with the chosen sample or to a maximum of 25 cycles, whichever comes first. | 45 devices |

1/ A separate sample may be pulled for each test.



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* TABLE III. Test ratings for diodes types 1N3821A through 1N3828A and 1N3016B through 1N3051B (5 percent tolerance).

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | Col 8 | Col 9 | Col 10 | Col 11 | Col 12 | | Col 13 | | Col 14 | Col 15 | Col 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Voltage Group 1/ 2/ | $V_Z$ Nom 2/ | $V_Z$ Min 2/ | $V_Z$ Max 2/ | $I_Z$ Test current | $Z_{ZT}$ Impedance | $Z_{ZK}$ Knee impedance | $I_{ZM}$ Max dc current | $V_z$ (reg) | $I_R$ (Surge) $T_A$ = +25°C | $V_R$ Reverse voltage | $I_{R1}$ Reverse current dc | | $I_{R3}$ Reverse current dc | | (-1 only) $I_{R2}$ Reverse current dc; $T_A$ = +150°C | $\alpha_{VZ}$ Temperature coefficient | $I_{zk}$ Test current |
| | Volts | Volts | Volts | mA | ohms | ohms | mA | Volts | mA | Volts | non -1 µA | -1 µA | non-1 µA | -1 µA | µA | %/°C | mA |
| 1N3821A | 3.3 | 3.14 | 3.46 | 76 | 10 | 400 | 276 | 1.00 | 1380 | 1 | 100 | 100 | 200 | 150 | 200 | -0.075 | 1.0 |
| 1N3822A | 3.6 | 3.42 | 3.78 | 69 | 10 | 400 | 252 | 0.80 | 1260 | 1 | 100 | 75 | 200 | 100 | 150 | - .070 | 1.0 |
| 1N3823A | 3.9 | 3.71 | 4.09 | 64 | 9 | 400 | 238 | 0.75 | 1190 | 1 | 50 | 25 | 100 | 40 | 100 | - .060 | 1.0 |
| 1N3824A | 4.3 | 4.09 | 4.51 | 58 | 9 | 400 | 213 | 0.70 | 1070 | 1 | 10 | 5 | 20 | 10 | 50 | - .050 | 1.0 |
| 1N3825A | 4.7 | 4.47 | 4.93 | 53 | 8 | 500 | 194 | 0.60 | 970 | 1 | 10 | 5 | 20 | 10 | 50 | ± .025 | 1.0 |
| 1N3826A | 5.1 | 4.85 | 5.35 | 49 | 7 | 550 | 178 | 0.50 | 890 | 1 | 10 | 3 | 20 | 6 | 50 | ± .030 | 1.0 |
| 1N3827A | 5.6 | 5.32 | 5.88 | 45 | 5 | 600 | 162 | 0.40 | 810 | 2 | 10 | 3 | 20 | 6 | 50 | + .040 | 1.0 |
| 1N3828A | 6.2 | 5.89 | 6.51 | 41 | 2 | 700 | 146 | 0.30 | 730 | 3 | 10 | 3 | 20 | 6 | 50 | + .050 | 1.0 |
| 1N3016B | 6.8 | 6.46 | 7.14 | 37 | 3.5 | 700 | 140 | 0.30 | 740 | 5.2 | 150 | 5.0 | 300 | 10 | 50 | + .057 | 1.0 |
| 1N3017B | 7.5 | 7.13 | 7.87 | 34 | 4.0 | 700 | 125 | 0.35 | 680 | 5.7 | 100 | 5.0 | 200 | 10 | 50 | + .061 | 0.5 |
| 1N3018B | 8.2 | 7.79 | 8.61 | 31 | 4.5 | 700 | 115 | 0.40 | 600 | 6.2 | 50 | 5.0 | 100 | 10 | 50 | + .065 | 0.5 |
| 1N3019B | 9.1 | 8.65 | 9.55 | 28 | 6.0 | 700 | 105 | 0.45 | 540 | 6.9 | 25 | 5.0 | 50 | 10 | 50 | + .068 | 0.5 |
| 1N3020B | 10 | 9.5 | 10.5 | 25 | 7 | 700 | 95 | 0.50 | 480 | 7.6 | 25 | 5.0 | 50 | 10 | 50 | + .071 | 0.25 |
| 1N3021B | 11 | 10.45 | 11.55 | 23 | 8 | 700 | 85 | 0.55 | 420 | 8.4 | 10 | 1.0 | 20 | 4 | 10 | + .073 | 0.25 |
| 1N3022B | 12 | 11.40 | 12.60 | 21 | 9 | 700 | 80 | 0.60 | 400 | 9.1 | 10 | 1.0 | 20 | 4 | 10 | + .076 | 0.25 |
| 1N3023B | 13 | 12.35 | 13.65 | 19 | 10 | 700 | 74 | 0.65 | 370 | 9.9 | 10 | 0.5 | 20 | 2 | 10 | + .079 | 0.25 |
| 1N3024B | 15 | 14.25 | 15.75 | 17 | 14 | 700 | 63 | 0.75 | 320 | 11.4 | 10 | 0.5 | 20 | 2 | 10 | + .082 | 0.25 |
| 1N3025B | 16 | 15.20 | 16.80 | 15.5 | 16 | 700 | 60 | 0.80 | 300 | 12.2 | 10 | 0.5 | 20 | 2 | 10 | + .083 | 0.25 |
| 1N3026B | 18 | 17.10 | 18.90 | 14.0 | 20 | 750 | 52 | 0.83 | 260 | 13.7 | 10 | 0.5 | 20 | 2 | 10 | + .085 | 0.25 |
| 1N3027B | 20 | 19.0 | 21.0 | 12.5 | 22 | 750 | 47 | 0.95 | 240 | 15.2 | 10 | 0.5 | 20 | 2 | 10 | + .086 | 0.25 |
| 1N3028B | 22 | 20.9 | 23.1 | 11.5 | 23 | 750 | 43 | 1.0 | 210 | 16.7 | 10 | 0.5 | 20 | 2 | 10 | + .087 | 0.25 |
| 1N3029B | 24 | 22.8 | 25.2 | 10.5 | 25 | 750 | 40 | 1.1 | 200 | 18.2 | 10 | 0.5 | 20 | 2 | 10 | + .088 | 0.25 |
| 1N3030B | 27 | 25.7 | 28.3 | 9.5 | 35 | 750 | 34 | 1.3 | 170 | 20.6 | 10 | 0.5 | 20 | 2 | 10 | + .090 | 0.25 |
| 1N3031B | 30 | 28.5 | 31.5 | 8.5 | 40 | 1000 | 31 | 1.4 | 160 | 22.8 | 10 | 0.5 | 20 | 2 | 10 | + .092 | 0.25 |
| 1N3032B | 33 | 31.4 | 34.6 | 7.5 | 45 | 1000 | 28 | 1.5 | 150 | 25.1 | 10 | 0.5 | 20 | 2 | 10 | + .091 | 0.25 |
| 1N3033B | 36 | 34.2 | 37.8 | 7.0 | 50 | 1000 | 26 | 1.7 | 130 | 27.4 | 10 | 0.5 | 20 | 2 | 10 | + .093 | 0.25 |

See footnotes at end of table.



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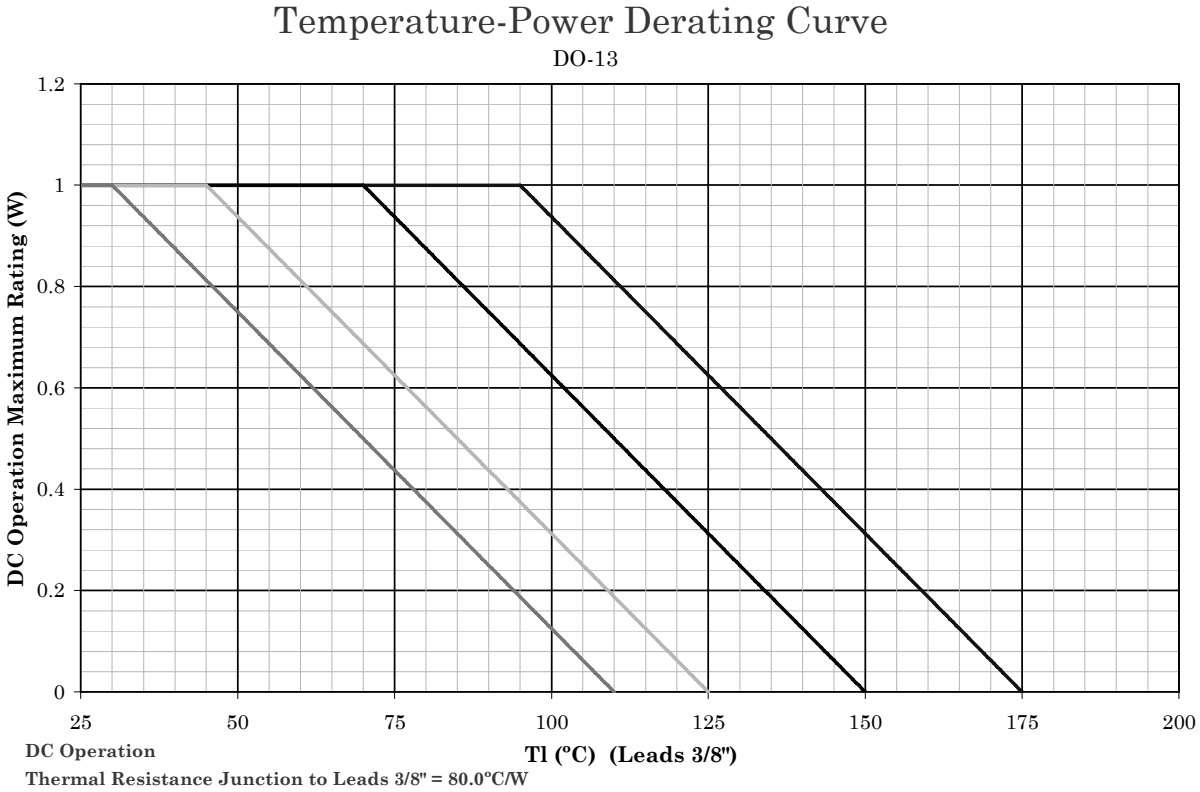
* TABLE III. Test ratings for diodes types 1N3821A through 1N3828A and 1N3016B through 1N3051B (5 percent tolerance).

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | Col 8 | Col 9 | Col 10 | Col 11 | Col 12 | | Col 13 | | Col 14 | Col 15 | Col 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Voltage Group 1/ 2/ | $V_Z$ Nom 2/ | $V_Z$ Min 2/ | $V_Z$ Max 2/ | $I_Z$ Test current | $Z_{ZT}$ Impedance | $Z_{ZK}$ Knee impedance | $I_{ZM}$ Max dc current | $V_z$ (reg) | $I_R$ (Surge) $T_A$ = +25°C | $V_R$ Reverse voltage | $I_{R1}$ Reverse current dc | | $I_{R3}$ Reverse current dc | | (-1 only) $I_{R2}$ Reverse current dc; $T_A$ = +150°C | $\alpha_{VZ}$ Temperature coefficient | $I_{zk}$ Test current |
| | Volts | Volts | Volts | mA | ohms | ohms | mA | Volts | mA | Volts | non -1 µA | -1 µA | non-1 µA | -1 µA | µA | %/°C | mA |
| 1N3034B | 39 | 37.1 | 40.9 | 6.5 | 60 | 1000 | 23 | 1.8 | 110 | 29.7 | 10 | 0.5 | 20 | 2 | 10 | + .094 | 0.25 |
| 1N3035B | 43 | 40.9 | 45.1 | 6.0 | 70 | 1500 | 21 | 1.9 | 100 | 32.7 | 10 | 0.5 | 20 | 2 | 10 | + .095 | 0.25 |
| 1N3036B | 47 | 44.7 | 49.3 | 5.5 | 80 | 1500 | 19 | 2.1 | 95 | 35.8 | 10 | 0.5 | 20 | 2 | 10 | + .095 | 0.25 |
| 1N3037B | 51 | 48.5 | 53.5 | 5.0 | 95 | 1500 | 18 | 2.3 | 90 | 38.8 | 10 | 0.5 | 20 | 2 | 10 | + .096 | 0.25 |
| 1N3038B | 56 | 53.2 | 58.8 | 4.5 | 110 | 2000 | 17 | 2.5 | 85 | 42.6 | 10 | 0.5 | 20 | 2 | 10 | + .096 | 0.25 |
| 1N3039B | 62 | 58.95 | 65.1 | 4.0 | 125 | 2000 | 15 | 2.7 | 75 | 47.1 | 10 | 0.5 | 20 | 2 | 10 | +0.097 | 0.25 |
| 1N3040B | 68 | 64.60 | 71.4 | 3.7 | 150 | 2000 | 14 | 3.0 | 70 | 51.7 | 10 | 0.5 | 20 | 2 | 10 | +0.097 | 0.25 |
| 1N3041B | 75 | 71.35 | 78.7 | 3.3 | 175 | 2000 | 12 | 3.3 | 63 | 56.0 | 10 | 0.5 | 20 | 2 | 10 | +0.098 | 0.25 |
| 1N3042B | 82 | 77.95 | 86.1 | 3.0 | 200 | 3000 | 11 | 3.6 | 58 | 62.2 | 10 | 0.5 | 20 | 2 | 10 | +0.098 | 0.25 |
| 1N3043B | 91 | 86.5 | 95.5 | 2.8 | 250 | 3000 | 10 | 4.0 | 50 | 69.2 | 10 | 0.5 | 20 | 2 | 10 | +0.099 | 0.25 |
| 1N3044B | 100 | 95.0 | 105.0 | 2.5 | 350 | 3000 | 9 | 4.4 | 45 | 76.0 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3045B | 110 | 104.5 | 115.5 | 2.3 | 450 | 4000 | 8.3 | 5.0 | 42 | 83.6 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3046B | 120 | 114 | 126 | 2.0 | 550 | 4500 | 8.0 | 5.5 | 40 | 91.2 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3047B | 130 | 123.5 | 136.5 | 1.9 | 700 | 5000 | 6.9 | 6.0 | 35 | 98.8 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3048B | 150 | 142.5 | 157.5 | 1.7 | 1000 | 6000 | 5.7 | 7.0 | 29 | 114.0 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3049B | 160 | 152 | 168 | 1.6 | 1100 | 6500 | 5.4 | 8.0 | 27 | 121.6 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3050B | 180 | 171 | 189 | 1.4 | 1200 | 7000 | 4.9 | 10.0 | 25 | 136.8 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |
| 1N3051B | 200 | 190 | 210 | 1.2 | 1500 | 8000 | 4.6 | 12.0 | 23 | 152.0 | 10 | 0.5 | 20 | 2 | 10 | +0.100 | 0.25 |

1/ Ratings also apply to dash one and surface mount devices unless otherwise noted.
2/ For 5 percent voltage, tolerances are shown in table.
For 2 percent tolerance (“C” suffix “-1” suffix and JANHC only), column 3 is 2 percent less than column 2, column 4 is 2 percent more than column 2.
For 1 percent tolerance (“D” suffix “-1” suffix and JANHC only), column 3 is 1 percent less than column 2, column 4 is 1 percent more than column 2.




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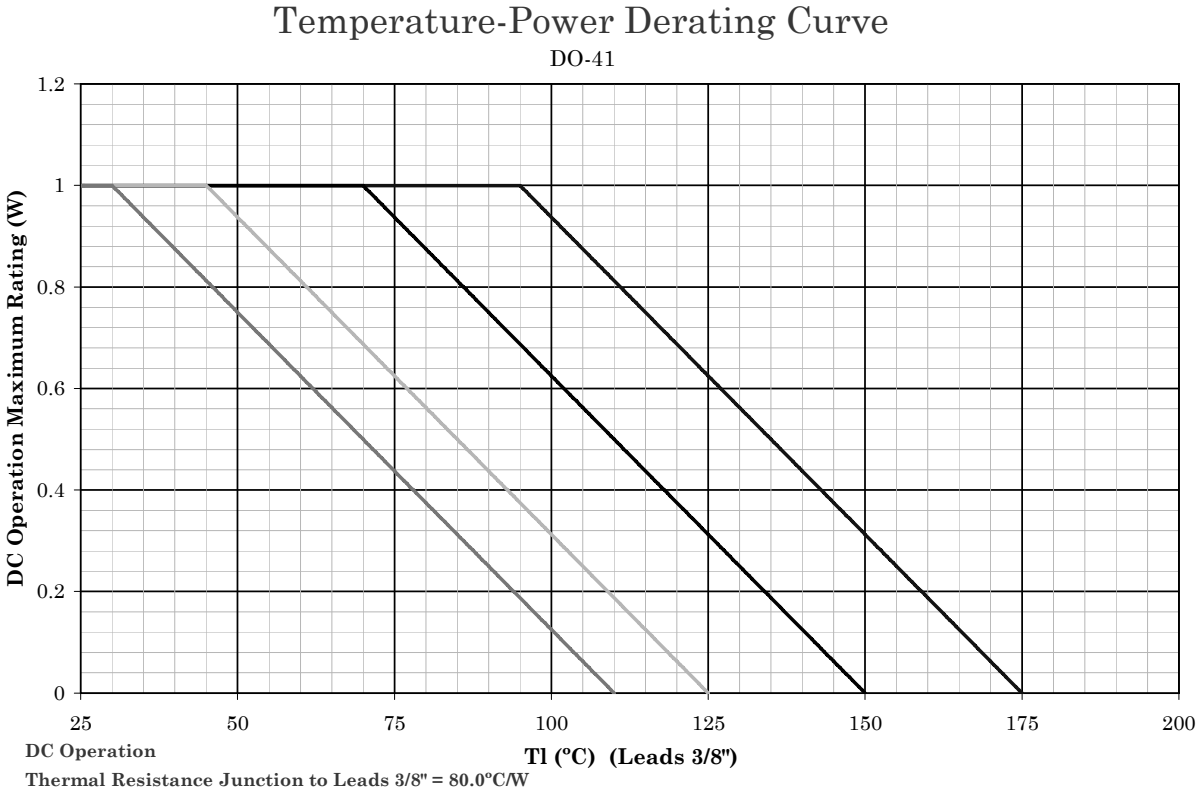

NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 175°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 6. Temperature-power derating curve (DO-13).



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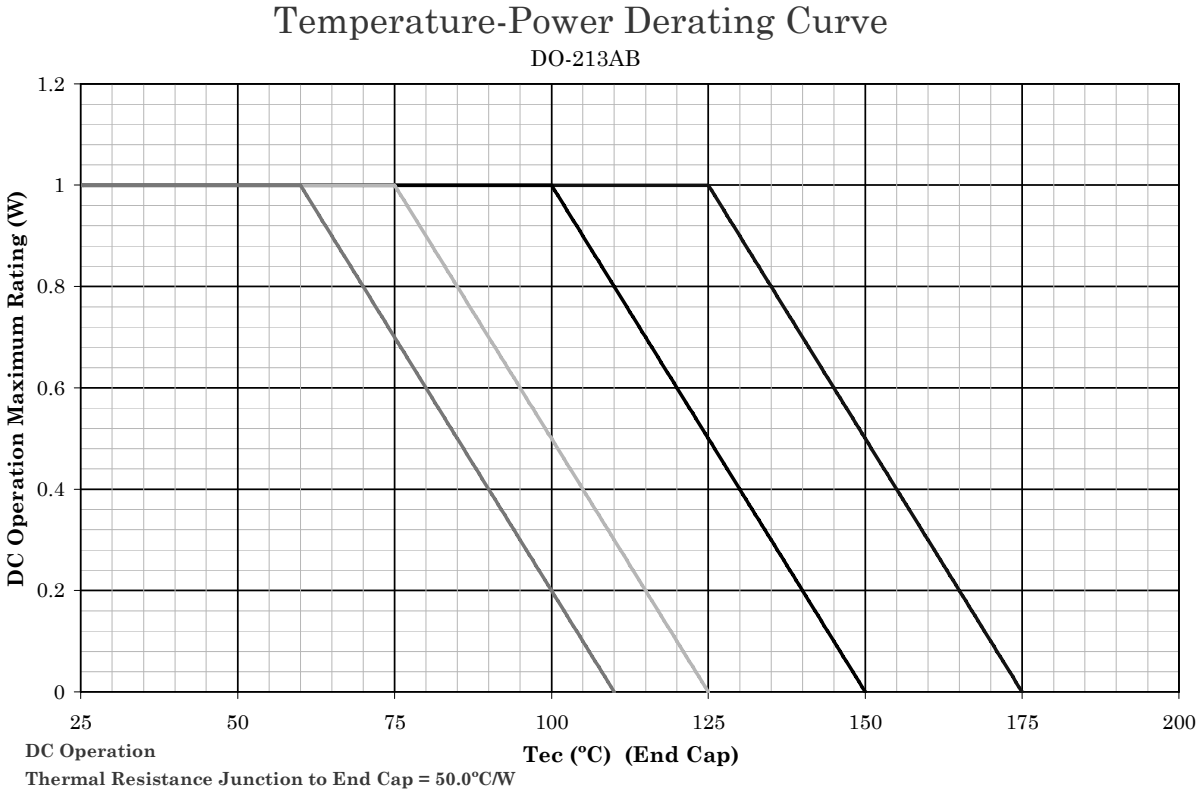

NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 175°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 7. Temperature-power derating curve (DO-41).



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NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 175°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 8. Temperature-power derating curve (DO-213AB).



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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil.

6.4 Substitution information. Device types 1N3821A through 1N3828A and 1N3016B through 1N3051B (excluding JANHC devices) are inactive for new design as of the date of this specification.

6.4.1 Substitutability of 2 percent and 1 percent tolerance devices. Devices of tighter tolerance are a direct one-way substitute for the looser tolerance devices (example: JANTX1N3821D-1 substitutes for JANTX1N3821A-1).



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* 6.4.2 Substitutability of dash one parts. Dash-one devices are a direct substitute for non dash-one devices and are preferred. The following table shows the direct substitutability.

| Superseded part number | Superseding part number | | Superseded part number | Superseding part number | | Superseded part number | Superseding part number |
|---|---|---|---|---|---|---|---|
| 1N3821A | 1N3821A-1 | | 1N3023B | 1N3023B-1 | | 1N3038B | 1N3038B-1 |
| 1N3822A | 1N3822A-1 | | 1N3024B | 1N3024B-1 | | 1N3039B | 1N3039B-1 |
| 1N3823A | 1N3823A-1 | | 1N3025B | 1N3025B-1 | | 1N3040B | 1N3040B-1 |
| 1N3824A | 1N3824A-1 | | 1N3026B | 1N3026B-1 | | 1N3041B | 1N3041B-1 |
| 1N3825A | 1N3825A-1 | | 1N3027B | 1N3027B-1 | | 1N3042B | 1N3042B-1 |
| 1N3826A | 1N3826A-1 | | 1N3028B | 1N3028B-1 | | 1N3043B | 1N3043B-1 |
| 1N3827A | 1N3827A-1 | | 1N3029B | 1N3029B-1 | | 1N3044B | 1N3044B-1 |
| 1N3828A | 1N3828A-1 | | 1N3030B | 1N3030B-1 | | 1N3045B | 1N3045B-1 |
| 1N3016B | 1N3016B-1 | | 1N3031B | 1N3031B-1 | | 1N3046B | 1N3046B-1 |
| 1N3017B | 1N3017B-1 | | 1N3032B | 1N3032B-1 | | 1N3047B | 1N3047B-1 |
| 1N3018B | 1N3018B-1 | | 1N3033B | 1N3033B-1 | | 1N3048B | 1N3048B-1 |
| 1N3019B | 1N3019B-1 | | 1N3034B | 1N3034B-1 | | 1N3049B | 1N3049B-1 |
| 1N3020B | 1N3020B-1 | | 1N3035B | 1N3035B-1 | | 1N3050B | 1N3050B-1 |
| 1N3021B | 1N3021B-1 | | 1N3036B | 1N3036B-1 | | 1N3051B | 1N3051B-1 |
| 1N3022B | 1N3022B-1 | | 1N3037B | 1N3037B-1 | | | |

6.5 Suppliers of JANHC die. The qualified JANHC suppliers with the applicable letter version (example JANHCA1N3821A) will be identified on the QML.

| JANHC ordering information | | |
|---|---|---|
| PIN | Manufacturer CAGE | |
| | 43611 | 12954 |
| 1N3821A through 1N3828A | JANHCA1N3821A through JANHCA1N3828A | JANHCB1N3821A through JANHCB1N3828A |
| 1N3016B through 1N3051B | JANHCA1N3016B through JANHCA1N3051B | JANHCB1N3016B through JANHCB1N3051B |

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA – CC

Preparing activity:
DLA - CC

(Project 5961-2007-122)

Review activities:
Army - AR, AV, MI, SM
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